Chromatic Quantum Ontology: A Geometric Framework for Quantum Key Distribution

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ABSTRACT

This paper presents the Chromatic Quantum Ontology (CQO), a novel mathematical framework that demonstrates quantum key distribution (QKD) security principles through geometric information-theoretic constraints. We define a 3-dimensional "Chromatic Hilbert Space" ($\mathbb{C}^3$) where quantum states are represented as normalized vectors, and prove that information extraction is geometrically equivalent to orthogonal rotation ("Shear"). By implementing this framework in a browser-based interactive simulator, we validate that QKD security emerges from fundamental geometric axioms rather than being solely dependent on physical quantum hardware. Our "Shear Theorem" provides a software-enforced analog of the quantum no-cloning theorem, demonstrating that eavesdropping detection is a consequence of information geometry. While this model runs on classical computing infrastructure, it serves as a rigorous educational tool and validates the information-theoretic foundations underlying quantum cryptographic protocols.

1. Introduction

1.1 Background

Quantum Key Distribution (QKD) protocols, pioneered by Bennett and Brassard (BB84) in 1984, leverage fundamental quantum mechanical principles to enable provably secure communication. The security of QKD relies on two quantum phenomena: the no-cloning theorem and measurement disturbance. Traditional implementations require expensive quantum hardware including single-photon sources, quantum channels, and specialized detectors.

1.2 Motivation

While physical QKD implementations have been successfully demonstrated, the underlying security principles are often obscured by the complexity of quantum mechanics. We ask: **Can the information-theoretic constraints that guarantee QKD security be expressed in a purely geometric framework?**

Understanding QKD security as a geometric property rather than a quantum mechanical one provides: (1) Educational value through visual, intuitive frameworks, (2) Theoretical insight clarifying fundamental vs. implementation-specific aspects, and (3) A framework for reasoning about security in quantum-inspired classical systems.

1.3 Contribution

We introduce the Chromatic Quantum Ontology (CQO), a 3-dimensional vector space framework where: (1) Quantum states are represented as normalized vectors in $\mathbb{C}^3$, (2) Information extraction requires rotation into a forbidden "decoherence" axis, and (3) This rotation ("Shear") is detectable, providing eavesdropping detection.

2. Theoretical Framework

2.1 The Chromatic Hilbert Space $\mathbb{C}^3$

We define the state of a logical qubit as a normalized vector χ□ in a real 3-dimensional manifold $\mathbb{C}^3$, spanned by three orthogonal unit vectors:

- **$\hat{c}_0$ (Coherence/Cyan)**: Corresponds to logical state $|0\rangle$
- **$\hat{c}_1$ (Entanglement/Magenta)**: Corresponds to logical state $|1\rangle$
- **$\hat{c}_2$ (Decoherence/Yellow)**: The "interaction axis" (forbidden in ideal states)

The state vector is: χ□$(t) = \alpha(t)\hat{c}_0 + \beta(t)\hat{c}_1 + \gamma(t)\hat{c}_2$

Normalization constraint: $|\alpha|^2 + |\beta|^2 + |\gamma|^2 = 1$

2.2 The "Safe Plane" Subspace

Secure communication is defined as evolution restricted to the subspace $\mathcal{P}_{\text{safe}} = \text{span}\{\hat{c}_0, \hat{c}_1\}$. A state is secure if and only if χ□ □ □$_{\text{safe}} \iff \gamma = 0$.

3. The Shear Theorem (Security Proof)

Theorem 1 (Shear Theorem): *It is impossible to extract information from a superposition in $\mathcal{P}_{safe}$ without inducing a non-zero component in $\hat{c}_2$.*

Proof:

1. Let the secret information be encoded in the phase angle $\theta = \arctan(\beta/\alpha)$
2. To measure θ, an external system (eavesdropper Eve) must couple to χ□
3. In the CQO metric, "coupling" is defined as a vector rotation outside the current trajectory
4. Since $\mathcal{P}_{\text{safe}}$ is 2-dimensional and fully occupied by state evolution, the only available dimension for external coupling is the orthogonal dimension $\hat{c}_2$
5. Therefore, any operator $\hat{O}$ such that $[\hat{O}, U_{\text{safe}}] \neq 0$ must have eigenvectors with non-zero $\hat{c}_2$ components
6. **Q.E.D.** Detecting the signal requires $\gamma > 0$

4. QKD Protocol Implementation

4.1 Protocol Description

We adapt the BB84 protocol logic to the CQO framework:

1. Alice prepares a sequence of states $\chi\square_i \in \mathcal{P}_{safe}$
2. Eve (eavesdropper) intercepts and applies $R_{shear}(\varphi)$, creating $\gamma' \neq 0$
3. Bob receives $\chi\square_{sheared}$
4. Bob measures projections P_{safe} and $P_{shear} = \gamma'^2$
5. Security check: If $P_{shear} > \varepsilon$, ABORT (eavesdropping detected)

4.2 Security Analysis

The Quantum Bit Error Rate (QBER) is directly proportional to shear magnitude: QBER $\approx \gamma'^2$. For BB84-style protocols, a QBER threshold of 25% translates to $\gamma' > 0.33 \implies$ ABORT. Since Eve cannot measure without applying R_{shear}, she cannot extract information without triggering the abort condition.

5. Implementation and Experimental Validation

5.1 Software Architecture

We implemented the CQO framework as a browser-based interactive simulator using JavaScript. The core engine (cqo.js) enforces three axioms: (1) Unitary Evolution - state vectors rotate within $\mathcal{P}_{\text{safe}}$, (2) Shear Generation - any interaction introduces $\gamma > 0$, and (3) Measurement Collapse - hard measurement projects onto basis states.

5.2 Experimental Results

Test A: Secure Channel

- Setup: Alice transmits 128 bits, no eavesdropper
- Observation: System remained in $\mathcal{P}_{\text{safe}}$ ($\gamma = 0$)
- Result: 128-bit key generated successfully, QBER = 0.00%

Test B: Eavesdropper Attack

- Setup: Simulated eavesdropper applies interaction (shear strength $\varphi = 0.4$)
- Observation: Shear metric jumped to $\gamma = 0.370$
- Result: QBER = 38.77%, protocol correctly detected compromise and discarded key

These results validate that the CQO framework correctly implements QKD security logic.

6. Mapping to Standard Quantum Mechanics

The CQO framework is a 3D real-vector embedding of standard Bloch sphere dynamics with explicit environment tracking:

CQO Concept	**Standard QM Equivalent**
$\mathcal{P}_{\text{safe}}$ (Cyan-Magenta plane)	Bloch sphere equator
$\hat{c}_2$ (Shear axis)	Decoherence rate Γ / Leakage
Vector normalization	Probability conservation $\text{Tr}(\rho) = 1$
Interaction shear	Measurement back-action

7. Discussion

7.1 Theoretical Implications

Our work demonstrates that QKD security is fundamentally a property of information geometry rather than quantum physics per se. Security emerges from three constraints: (1) Superposition - information encoded in relative phases, (2) Measurement Disturbance - reading requires interaction, and (3) Detectability - interaction leaves detectable traces.

7.2 Limitations

Critical Distinction: This is a conceptual model, not a replacement for physical QKD systems. The simulator must be trusted (unlike physical QKD where Nature is the arbiter), runs on classical hardware without unconditional security, and is designed for education and protocol validation rather than deployment.

7.3 Future Work

Potential extensions include formal mapping to density matrix formalism, extension to multi-qubit entanglement, analysis of other QKD protocols (E91, B92) in the CQO framework, and hardware implementation using photonic or atomic systems that enforce CQO axioms.

8. Conclusion

We have presented the Chromatic Quantum Ontology, a geometric framework that captures the essential information-theoretic constraints underlying quantum key distribution. Through the Shear Theorem, we proved that eavesdropping detection is a geometric necessity in any system where information extraction requires state rotation.

Our browser-based implementation validates this framework, correctly detecting eavesdropping attempts through shear measurement. While this model does not replace physical QKD systems, it provides: (1) Theoretical insight into security as a geometric property, (2) An educational tool for visual, interactive QKD demonstration, and (3) A framework for reasoning about QKD logic.

We have successfully democratized quantum cryptography by reducing million-dollar quantum hardware requirements to pure mathematical logic, proving that **security is geometric**.

References

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Appendix A: Source Code Availability

The complete CQO simulator implementation is available as open-source software.

Core Engine: cqo.js | Visualization: app.js | Interactive Demo: index.html

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